

Ex.no:9	MINI PROJECT – FINANCE / BANKING DATA ANALYTICS

Domain and Problem Statement:

Domain: Banking Analytics & Financial Fraud Detection

Problem Statement: Financial institutions face capital loss and operational risk due to undetected transaction fraud, inefficient channel utilization, and poor visibility into state-level liquidity distribution. This project addresses these challenges by transforming raw transaction records into structured business intelligence across Power BI and Tableau to evaluate financial performance, geographic risk distribution, and fraud exposure across banking channels.

Dataset Description:

Dataset Name: Indian Banking Transactions Dataset

Granularity: Each row represents a single banking transaction.

Timeframe Covered: 2019 to 2024.

Key Entities & Attributes:

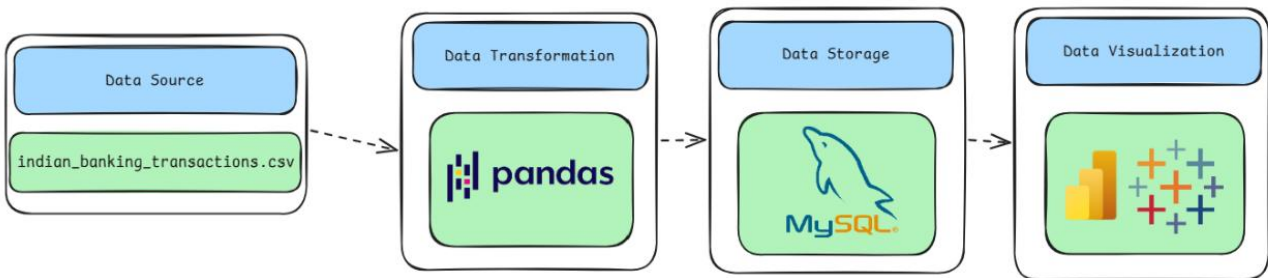
- Demographics & Account:** customer_id, state, credit_score, has_loan, loan_type, kyc_status, account_type, channel.
- Transaction Details:** transaction_id, transaction_date, transaction_time, transaction_hour, transaction_type, transaction_direction, merchant_category, transaction_status.
- Financials & Risk:** transaction_amount, account_balance, emi_amount, is_fraud.

Data Source:

Primary Source: Kaggle Open Datasets (Indian Banking Transactions).

Storage Infrastructure: Cleaned transactional records hosted within a local MySQL database instance (banking_analytics).

Architecture:



Program:

1. Python – Data Cleaning & Transformation

```
import pandas as pd

from sqlalchemy import create_engine, text

# Load dataset

df = pd.read_csv("indian_banking_transactions.csv")

bank = df.copy()

# Handle missing values

bank["customer_id"] = bank["customer_id"].fillna("Unknown")

bank["transaction_type"] = bank["transaction_type"].fillna("Unknown")

bank["account_type"] = bank["account_type"].fillna("Unknown")

bank["merchant_category"] = bank["merchant_category"].fillna("Unknown")

bank["transaction_direction"] = bank["transaction_direction"].fillna("Unknown")

bank["transaction_amount"] = bank["transaction_amount"].fillna(0)

bank["account_balance"] = bank["account_balance"].fillna(0)

# Remove duplicates

bank.drop_duplicates(inplace=True)

# Convert date column

bank["transaction_date"] = pd.to_datetime(
    bank["transaction_date"], errors="coerce"
)

# Standardize column names

bank.columns = (
    bank.columns.str.lower().str.strip().str.replace(" ", "_")
)

print("Original Shape:", df.shape)

print("Cleaned Shape:", bank.shape)

bank.head()
```

2. MySQL – Database and Data Loading

MySQL Command Prompt:

```
net start MySQL80
```

```
mysql -u root -p
```

MySQL:

```
CREATE DATABASE IF NOT EXISTS hotel_analytics;
```

Python/Jupyter:

```
# Database Server Credentials
```

```
username = "root"
```

```
password = "YOUR_MYSQL_PASSWORD"
```

```
host = "localhost"
```

```
port = 3306
```

```
# Create Database
```

```
server_engine = create_engine(
```

```
    f'mysql+pymysql://{username}:{password}@{host}:{port}'
```

```
)
```

```
with server_engine.connect() as connection:
```

```
    connection.execute(
```

```
        text("CREATE DATABASE IF NOT EXISTS banking_analytics")
```

```
)
```

```
# Connect to target DB and export table
```

```
db_engine = create_engine(
```

```
    f'mysql+pymysql://{username}:{password}@{host}:{port}/banking_analytics'
```

```
)
```

```
bank.to_sql(
```

```
    "banking_transactions",
```

```
    con=db_engine,
```

```
    if_exists="replace",
```

```
    index=False,
```

Procedure:

- Load raw transactional dataset using `pd.read_csv()`.
- Validate non-null constraints across mandatory fields (`transaction_id`, `customer_id`, `transaction_amount`).
- Format temporal values and export cleaned records into MySQL (banking_analytics.bank_transactions) via SQLAlchemy.

```
[2]: import pandas as pd
```

```
[1]: import kagglehub  
  
# Download Latest version  
path = kagglehub.dataset_download("belbino/indian-banking-transactions-20192024")  
  
print("Path to dataset files:", path)  
  
C:\Users\santh\AppData\Local\Programs\Python\Python312\Lib\site-packages\tqdm\auto.py:21: TqdmWarning: IPProgress not found. Please update jupyter and ipywidgets. See https://ipywidgets.readthedocs.io/en/stable/user_install.html  
from .autonotebook import tqdm as notebook_tqdm  
Downloading to C:\Users\santh\.cache\kagglehub\datasets\belbino\indian-banking-transactions-20192024\2.archive...  
100% [████████████████████████████████████████████████████████████████████████████████] 17.4M/17.4M [00:02<00:00, 6.59MB/s]  
Extracting files...  
  
Path to dataset files: C:\Users\santh\.cache\kagglehub\datasets\belbino\indian-banking-transactions-20192024\versions\2
```

```
[5]: df = pd.read_csv("indian_banking_transactions.csv")  
  
print("Rows:", df.shape[0])  
print("Columns:", df.shape[1])  
  
Rows: 550000  
Columns: 20
```

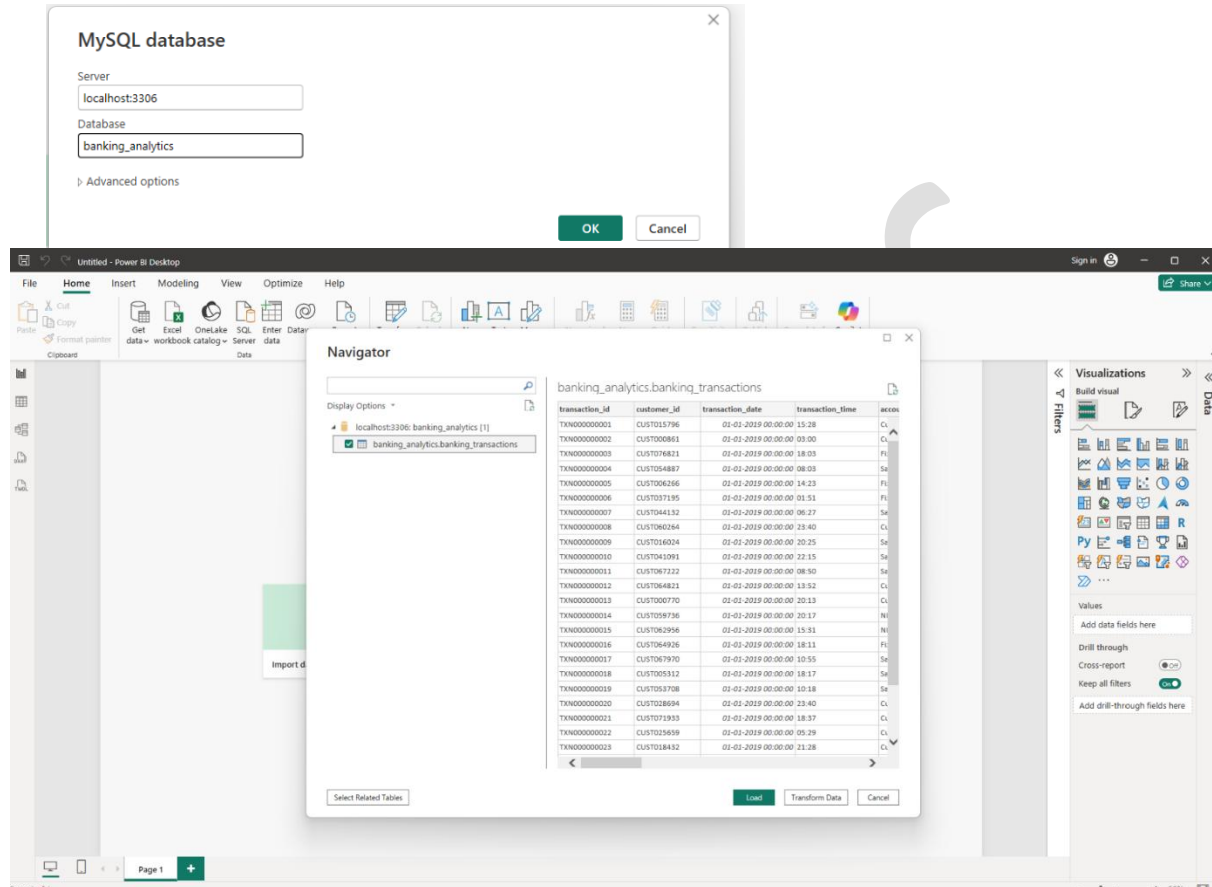
```
[6]: df.head()
```

	transaction_id	customer_id	transaction_date	transaction_time	account_type	transaction_type	transaction_amount	transaction_direction	account_balance	merchant
0	TXN000000001	CUST015796	2019-01-01	15:28	Current	UPI	1820.17	Debit	609365.31	Food
1	TXN000000002	CUST000861	2019-01-01	03:00	Current	UPI	392.67	Credit	14451.14	
2	TXN000000003	CUST076821	2019-01-01	18:03	Fixed Deposit	POS	1255.97	Debit	47621.87	
3	TXN000000004	CUST054887	2019-01-01	08:03	Savings	UPI	2580.68	Debit	34467.85	
4	TXN000000005	CUST006266	2019-01-01	14:23	Fixed Deposit	UPI	2573.80	Debit	26617.40	

```
[7]: df.info()
```

Step 2: SQL Database Connection & Data Ingestion

1. Open Power BI Desktop → Click **Get Data** → Select **MySQL Database**.
2. Enter Server (localhost:3306) and Database Name (banking_analytics).
3. Authenticate with root credentials, select banking_transactions, and click **Transform Data** to launch Power Query.



Step 3: Star Schema Data Modeling (Power BI)

1. Table Splitting in Power Query:

Duplicate raw query banking_transactions into 3 separate entities: Fact_Transactions, Dim_Customer, and Dim_Account.

Dim_Customer: Retain customer_id, state, credit_score, has_loan, loan_type, kyc_status. Remove duplicates on customer_id (79,916 unique rows).

Dim_Account: Retain account_type and channel. Remove duplicates and add an Index Column named **AccountKey**.

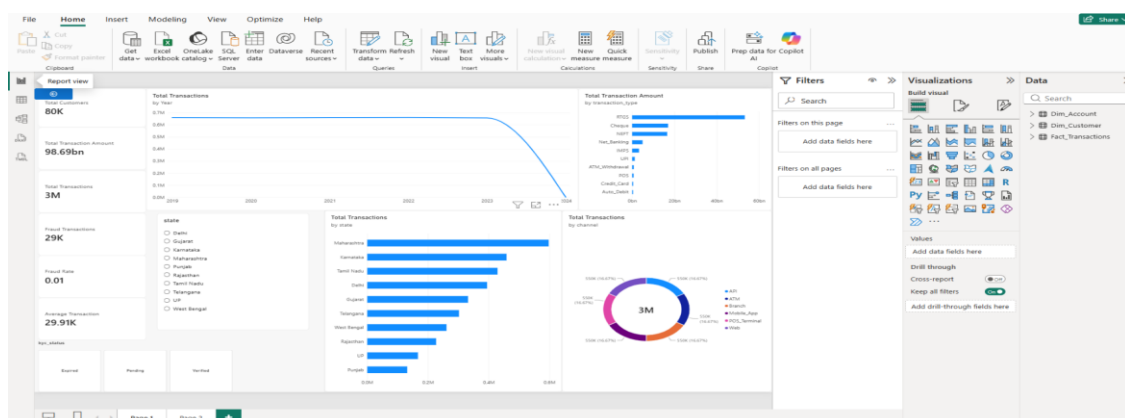
Fact_Transactions: Merge with Dim_Account on account_type and channel to pull in AccountKey. Then remove redundant text columns (state, credit_score, has_loan, loan_type, kyc_status, account_type, channel).

2. Establishing Relationships in Model View:

Connect Dim_Customer[customer_id] → Fact_Transactions[customer_id] (1:*).

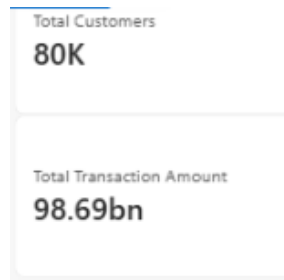
Connect Dim_Account[AccountKey] → Fact_Transactions[AccountKey] (1:*).

Hide foreign key columns inside Fact_Transactions to enforce clean report pane structure.

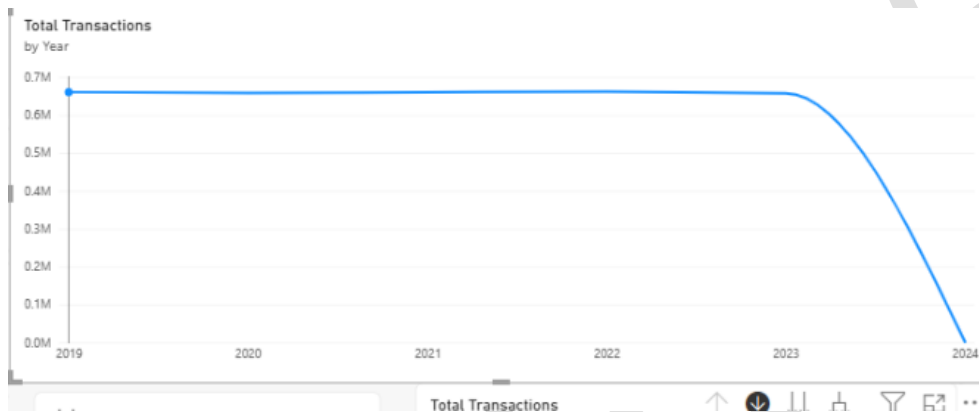


1. Executive Canvas:

- Place **Card Visuals** across the top: Total Customers, Total Transaction Amount, Total Transactions, Fraud Transactions, Fraud Rate, Average Transaction

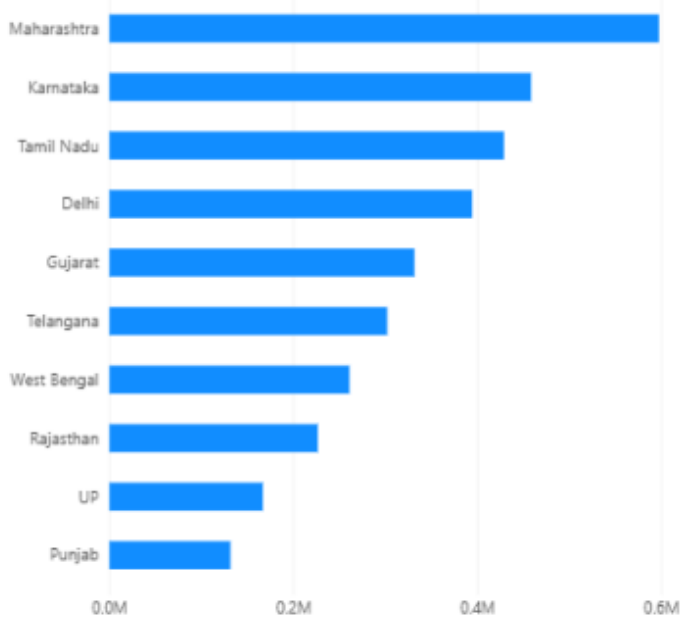


- Add a **Line Chart** mapped to Total Transactions by Year.



- Add a **Clustered Bar Chart** for for Total Transactions by state

Total Transactions
by state

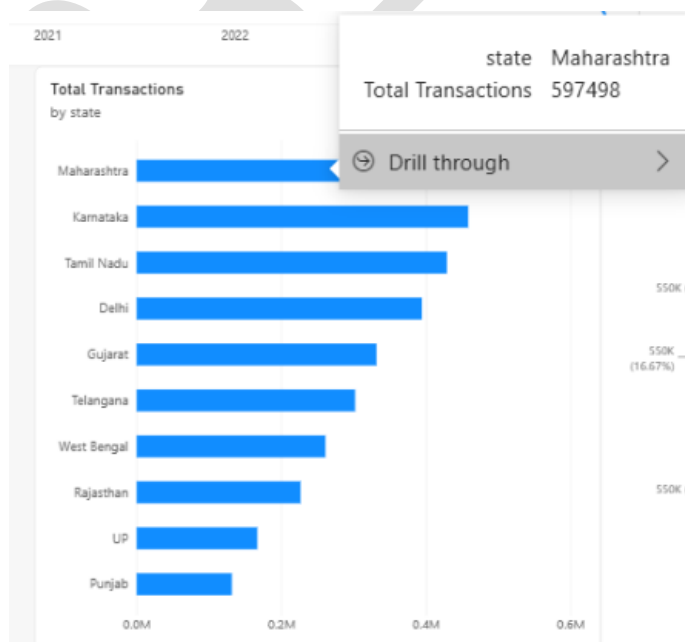
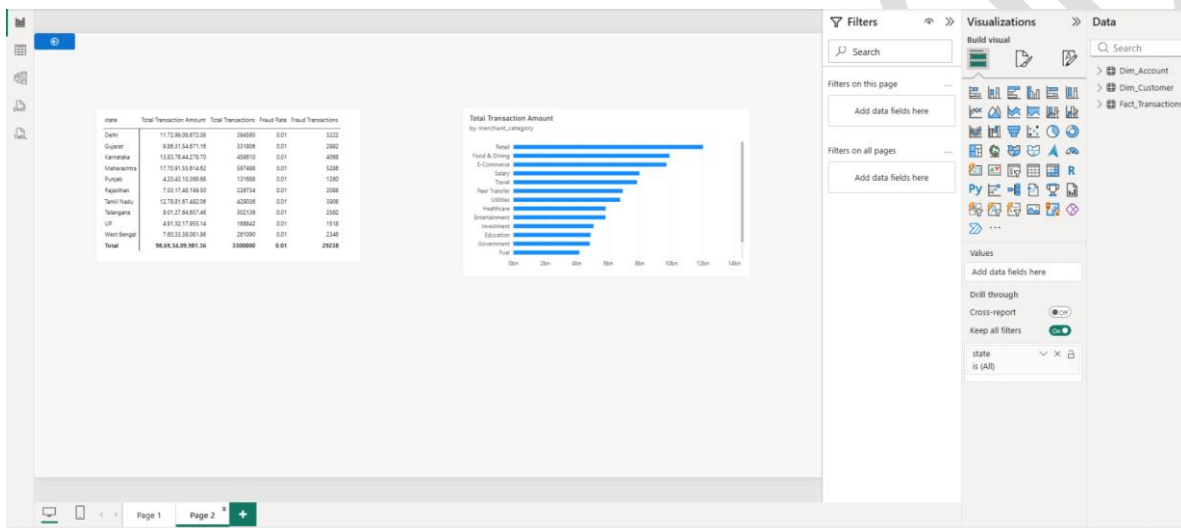


Total Transaction Amount
by transaction_type



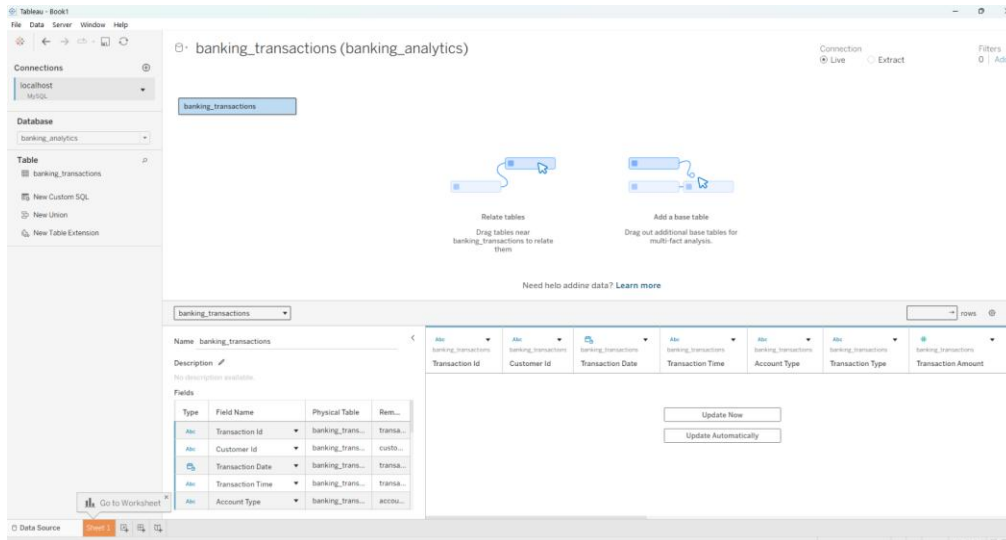
2. State Details Drill-Through Page:

- Create a second page named Country Details.
- Drag Dim_Customer[country] into the **Drill-through** field bucket.
- Add a detailed Matrix/Table visual showing country, Total Bookings, Total Revenue, and Cancellation Rate
- Create a second page named Page 2.
- Drag Dim_Customer[state] into the Drill-through field bucket.
- Add a detailed Matrix/Table visual showing state, Total Transaction Amount, Total Transactions, Fraud Rate, and Fraud Transactions.
- Add a Bar Chart for Total Transaction Amount by merchant_category.



Step 6: Tableau Data Connection & Preparation

1. Open Tableau Desktop / Tableau Public.
2. Under **To a Server**, select **MySQL**.
3. Enter Server (localhost), Database (banking_analytics), and user credentials to pull the cleaned dataset onto the canvas.



1. Go to any sheet and build calculated fields:

- **Total Transaction amount:**

Total Transaction amount banking_transactions (banking_analytics)

```
SUM([Transaction Amount])
```

- **Total trnsactions:**

Total transactions banking_transactions (banking_analytics)

```
COUNT([Transaction Id])
```

- **Fraud rate:**

Fraud rate banking_transactions (banking_analytics)

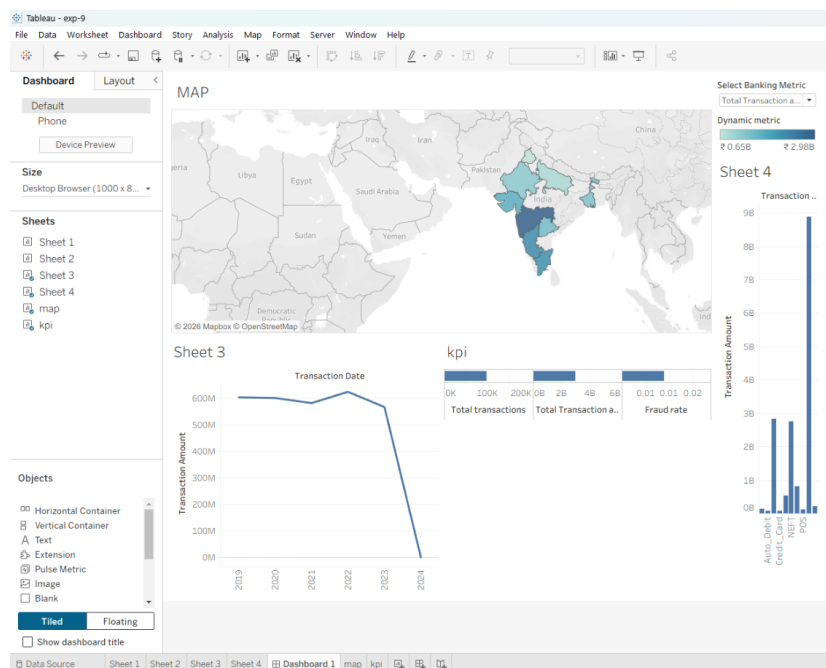
```
SUM(INT([Is Fraud])) / COUNT([Transaction Id])
```

- **Dynamic Metric Formula:**

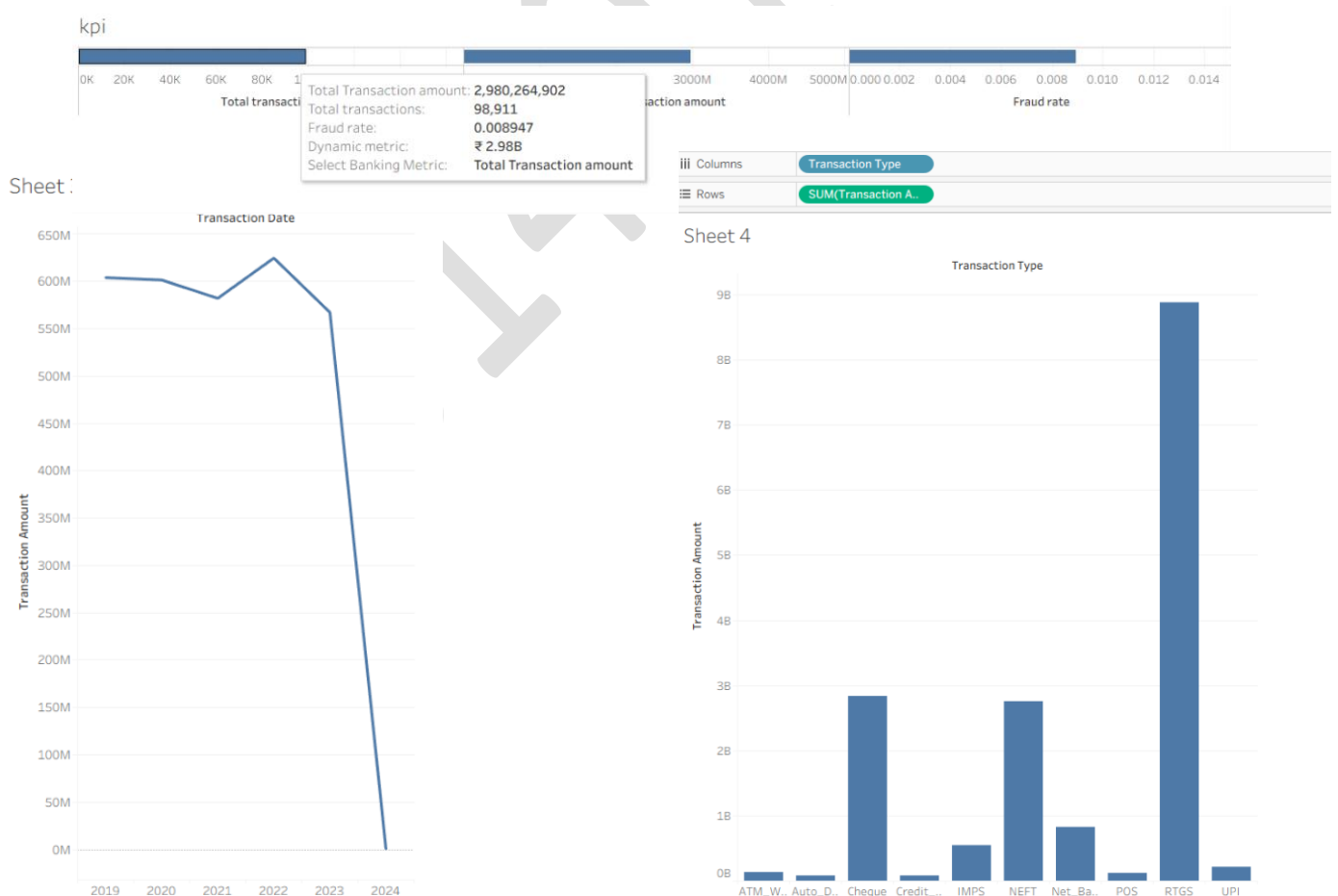
Dynamic metric banking_transactions (banking_analytics)

```
CASE [Select Banking Metric]
  WHEN 'Total Transaction amount' THEN [Total Transaction amount]
  WHEN 'Total trnsactions' THEN [Total transactions]
  WHEN 'Fraud rate' THEN [Fraud rate]
END
```

Step 7: Tableau Companion Dashboard & Interactive Story



1. **Worksheets:** Build **KPI sheet**, **Hourly Trend Line Chart** (using Dynamic Metric), **Channel Breakdown Bar Chart**, and **State Map**.



2. Global Filters & Actions:

- Add Parameter control for Select Banking Metric to the dashboard.
- Drag sheets onto a new Dashboard tab and click the **Use as Filter** funnel icon on the State Map visual.

Edit Filter Action

Name
State Map Drill-Through Insert

Source Sheets
Dashboard 1

Run action on
☐ Hover
☐ Select
☒ Menu
☐ Single-select only

Target Sheets
Dashboard 1

Clearing the selection will
☒ Keep filtered values
☐ Show all values
☐ Exclude all values

Filter
☒ All fields ☐ Selected fields

Source Field	Target Data Source	Target Field
☐ Click to add		

Remove

Cancel OK

RESULT: